An Energy-Based View on the Manifold Hypothesis

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Abstract

A commonly adopted assumption in modern machine learning is the *manifold hypothesis*, which claims that although data is embedded in a high-dimensional ambient space, the probability mass concentrates near a much lower-dimensional geometric structure. However, the real-world latent variables that images are drawn from are continuous, not discrete, and therefore cannot be completely captured by a finite manifold. Moreover, the assumption does not properly capture the union of manifolds, stratified spaces, and manifolds with singularities and boundaries. Drawing upon recent advances in energy-based models, we provide an energy-based perspective on the real-world latents, describing an Occam's razor style analog for the manifold hypothesis, which is more amenable to recent literature in the field. We theoretically prove that our construction recovers the manifold hypothesis for coarse functions, and validate it with numerical examples.

1 Introduction

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1.1 Related Works

NERF ??????????

Energy-based models ??????

Compositional generation ??????

Occam's razor ??

2 Problem Setting

Supervised Learning 101. Observing data $D = \{(x_i, y_i)\}_{i=1}^n$ we want to learn a deterministic function $f^* : \mathcal{X} \to \mathbb{R}$ such that $f^*(x_i) = y_i$ for all $i = 1, \dots, n$. Usually, we know the input space $\mathcal{X}$ via some base distribution ν_0 on $\mathcal{X}$, for instance, uniformly distribution over a compact set $\mathcal{X} \subset \mathbb{R}^d$. Moreover, we assume that smoother or more predictable the function f^* is, the better it generalizes to new data.

Notation. We refer to a metric space (X, d) as a Polish space if it is complete (every Cauchy sequence in (X, d) converges to a point in X) and separable (there exists a countable dense subset $D \subset X$). For instance, $\mathbb{R}^n$ with the Euclidean metric, separable Banach spaces, Hilbert spaces, and function spaces such as $L^2(M)$ or $H^s(M)$ (when M is a compact manifold) are all Polish spaces. Moreover, if X is Polish, then $(X, \mathcal{B}(X))$ enjoys strong regularity properties that make it a canonical measurable structure in probability theory, where $\mathcal{B}(X)$ is the Borel σ-algebra on X.

Definition 2.1 (Symmetric Dirichlet Space)**.** Let $(X, \mathcal{F}, \mu)$ be a σ-finite measure space. A *Dirichlet form* on $L^2(\mu)$ is a pair $(\mathcal{E}, \mathcal{D}(\mathcal{E}))$ such that:

1. $\mathcal{D}(\mathcal{E}) \subset L^2(\mu)$ is a dense linear subspace;
2. $\mathcal{E} : \mathcal{D}(\mathcal{E}) \times \mathcal{D}(\mathcal{E}) \to \mathbb{R}$ is a symmetric bilinear form;
3. (Non-negativity)
$$\mathcal{E}(f, f) \geq 0 \qquad \forall f \in \mathcal{D}(\mathcal{E});$$
4. (Closedness) $\mathcal{D}(\mathcal{E})$ is complete with respect to the norm
$$\|f\|_{\mathcal{E}}^2 := \|f\|_{L^2(\mu)}^2 + \mathcal{E}(f, f);$$
5. (Markov property) for every $f \in \mathcal{D}(\mathcal{E})$, the truncation $\widetilde{f} := (0 \vee f) \wedge 1$ satisfies
$$\widetilde{f} \in \mathcal{D}(\mathcal{E}) \quad \text{and} \quad \mathcal{E}(\widetilde{f}, \widetilde{f}) \leq \mathcal{E}(f, f).$$

The triple $(X, \mathcal{F}, \mu, \mathcal{E})$ is called a *Symmetric Dirichlet Space*.

The Dirichlet form encodes the geometry of the learning space via the energy functional: $\mathcal{E}(f, f)$ is the energy of the function f. Enforcing the Markov property allows us to define stochastic processes on the learning space: semi-group generated by $\mathcal{E}$ is a Markov semigroup.

The main idea of learning is that if $x_1 \approx x_2$ then $f(x_1) \approx f(x_2)$ for a function f. Dirichlet form enforces precisely this property. As it encodes the geometry of the learning space, it allows us to define a generalized Laplacian L on $L^2(\mu)$ and Sobolev classes of functions on X.

Definition 2.2 (Generalized Laplacian)**.** Let $(X, \mathcal{F}, \mu, \mathcal{E})$ be a measure space endowed with a Dirichlet form. There exists a unique self-adjoint operator L on $L^2(\mu)$ such that
$$\mathcal{E}(f, g) = \langle f, Lg \rangle_{L^2(\mu)}, \qquad \forall\, f \in \mathcal{D}(\mathcal{E}),\ g \in \mathcal{D}(L).$$
The operator L admits a spectral decomposition $L\phi_j = \lambda_j \phi_j$, with
$$0 = \lambda_0 \leq \lambda_1 \leq \lambda_2 \leq \cdots, \qquad \langle \phi_j, \phi_k \rangle_{L^2(\mu)} = \delta_{jk}.$$

The Markov property of the Dirichlet form $\mathcal{E}$ implies that e^{tL} is a Markov semigroup. Hence, our distribution μ is a stable state of the system: given a base distribution ν_0 on $\mathcal{X}$, the distribution μ is the limit of the Markov process $e^{tL}\nu_0$ as $t \to \infty$.

Definition 2.3 (Effective dimension)**.** The *effective dimension* of the learning space $(X, \mathcal{F}, \mu, \mathcal{E})$ is the quantity $d_{\text{eff}} \in (0, \infty)$ such that the eigenvalues of the corresponding Laplacian operator $L : L^2(\mu) \to L^2(\mu)$ satisfy
$$\lambda_j \asymp j^{2/d_{\text{eff}}} \qquad \text{as } j \to \infty,$$
or equivalently, the eigenvalue counting function satisfies
$$N(\lambda) := \#\{j : \lambda_j \leq \lambda\} \asymp \lambda^{d_{\text{eff}}/2} \qquad \text{as } \lambda \to \infty.$$

Or, more generally
$$\operatorname{tr} e^{tL} \sim t^{-d_{\text{eff}}/2} \quad \text{as } t \to 0^+$$

Definition 2.4 (Spectral Sobolev class)**.** For smoothness $s > 0$ and radius $R > 0$, the *spectral Sobolev class* of order s is

$$\mathcal{H}^s_\mu(R) := \Big\{ f \in L^2(\mu) : \|f\|^2_{\mathcal{H}^s} := \sum_{j\geq 1} \lambda_j^s \widehat{f}_j^{\,2} \leq R^2 \Big\}, \qquad \widehat{f}_j := \langle f, \phi_j \rangle_{L^2(\mu)}.$$

The condition $f \in \mathcal{H}^s_\mu(R)$ requires the spectral coefficients to decay as $|\widehat{f}_j| \leq R\, \lambda_j^{-s/2} \asymp R\, j^{-s/d_{\mathrm{eff}}}$: higher smoothness s enforces faster decay and hence greater regularity with respect to the geometry encoded by $\mathcal{E}$.

Definition 2.5 (Heat semigroup Besov space)**.** Let $P_t = e^{-tL}$ be the Markov semigroup generated by L. For $s > 0$ and $1 \leq p, q \leq \infty$, define

$$\|f\|_{B^s_{p,q}(L)} = \|f\|_{L^p(\mu)} + \left(\int_0^1 \Big(t^{-s/2} \|(I - P_t) f\|_{L^p(\mu)} \Big)^q \frac{dt}{t} \right)^{1/q}.$$

Then $B^s_{p,q}(L)$ consists of all f for which this norm is finite.

Definition 2.6 (Spectral wavelets associated to L)**.** Let $\varphi_0 \in C_c^\infty([0,\infty))$ satisfy $\varphi_0(\lambda) = 1$ for $\lambda \in [0,1]$ and $\operatorname{supp} \varphi_0 \subset [0,2]$. Define the dyadic wavelet kernel

$$\psi(\lambda) = \varphi_0(\lambda) - \varphi_0(2\lambda), \qquad \psi_j(\lambda) = \psi(2^{-j}\lambda), \quad j \geq 1,$$

so that $\{\varphi_0, \psi_j\}_{j\geq 1}$ forms a partition of unity:

$$\varphi_0(\lambda) + \sum_{j=1}^\infty \psi_j(\lambda) = 1 \qquad \text{for all } \lambda \geq 0.$$

The *wavelet blocks* are the operators

$$\Psi_0 = \varphi_0(L), \qquad \Psi_j = \psi_j(L) = \psi(2^{-j}L), \quad j \geq 1,$$

defined via the spectral calculus of L, yielding the Littlewood–Paley decomposition

$$f = \Psi_0 f + \sum_{j=1}^\infty \Psi_j f.$$

If L has discrete spectrum $0 \leq \lambda_1 \leq \lambda_2 \leq \cdots$ with eigenfunctions $\{e_k\}$, then

$$\Psi_j f = \sum_k \psi_j(\lambda_k)\, \langle f, e_k \rangle_{L^2(\mu)}\, e_k.$$

3 Manifold Hypothesis

A commonly adopted assumption in modern machine learning is the *manifold hypothesis*, which claims that although data is embedded in a high-dimensional ambient space, the probability mass concentrates near a much lower-dimensional geometric structure.

Assumption 3.1 (Manifold Hypothesis)**.** Let $\chi \subset \mathbb{R}^D$ be the ambient space and let $\mathcal{D}$ be a probability distribution on χ. There exists a d-dimensional manifold $\mathcal{M} \subset \chi$, with $d \ll D$, such that

$$\mathbb{P}_{x\sim\mathcal{D}}\big(\operatorname{dist}(x, \mathcal{M}) \leq \varepsilon\big) \geq 1 - \delta,$$

for sufficiently small $\varepsilon, \delta > 0$.

In practice, data rarely concentrate on a single smooth manifold. Rather, they are better modeled as lying on *unions of manifolds*, or more generally on *stratified spaces* composed of pieces of varying intrinsic dimensions, potentially with singularities and boundaries.

Why small d is important?

- Generalization error depends on the intrinsic dimension d of the manifold.
- The number of samples n needed to learn the manifold increases with d.
- The complexity of the model increases with d.
- Can get a small embedding.

Shortcomings

- Doesn't properly capture union of manifolds.
- Doesn't properly capture stratified spaces.
- Doesn't properly capture manifolds with singularities and boundaries.
- What about infinite-dimensional data? Cause images come from infinite-dimensional continuous functions.

[Emile says: would be nice to elaborate on the above, and to point to specific instances where assuming it has some "catastrophic" consequences. that could go into a motivating examples section]

Example: Vision Data Each 2d image (or 3d scene or a point cloud) can be modeled $f : \mathbb{R}^2 \to \mathbb{R}^3$ as a function from coordinates to RGB (or XYZ) values. That's how NeRFS work:

Neural Radiance Fields (NeRFs). [Emile says: this actually looks very relevant. would be worth doing a deeper dive into this] A *Neural Radiance Field* models a 3D scene as a continuous function

$$f_\theta : (\mathbf{x}, \mathbf{d}) \;\mapsto\; (\sigma, \mathbf{c}),$$

where

- $\mathbf{x} \in \mathbb{R}^3$ denotes a spatial location,
- $\mathbf{d} \in \mathbb{S}^2$ denotes a viewing direction,
- $\sigma \in \mathbb{R}_{\geq 0}$ is the volume density at $\mathbf{x}$,
- $\mathbf{c} \in \mathbb{R}^3$ is the emitted RGB color in direction $\mathbf{d}$.

The key idea is that the scene is represented neither as a mesh, nor as a voxel grid, nor as a point cloud, but as a *continuous radiance field* parameterized by a neural network.

In a similar way, each 2d image can be modeled as a function from coordinates to RGB values: $f : \mathbb{R}^2 \to \mathbb{R}^3$ and the embedding of this function would the weights of the corresponding ReLU network. The regular manifold hypothesis doesn't properly capture this structure that:

- In a union of manifolds
- Has some hierarchical structure.
- Has permutation-invariance with respect to weights in each layer.

Also, importantly, each image f is not a smooth function: its a combination of a "simple" in some sense partition of the image into smooth enough regions.

4 Effective Dimension Hypothesis

Let us formulate the extension of the manifold hypothesis to this case. [Emile says: more generally, need to define energy functions, why they're relevant to this problem, and what characteristics do we want our manifold hypothesis to have?]

Assumption 4.1 (Effective Dimension Hypothesis)**.** All the observed data x_i come from a learning space $(X, \mathcal{F}, \mu, \mathcal{E})$ with small effective dimension even if the ambient space $(X, \mathcal{F}, \nu_0, \mathcal{E})$ has large effective dimension where ν_0 is the base distribution on $\mathcal{X}$:

$$d_{\text{eff}}(X, \mathcal{F}, \mu, \mathcal{E}) \ll d_{\text{eff}}(X, \mathcal{F}, \nu_0, \mathcal{E})$$

This assumption ensures two things: existence of the efficient coordinate system in the space $\mathcal{X}$ and efficient learning rate for smooth functions. Moreover, this assumption of the underlying learning space is strictly stronger than the manifold hypothesis and unifies multiple learning problems, e.g. classical kernel methods, flow matching/diffusion models and transformers.

4.1 Generalization Error

Proposition 4.2 (Bias-variance tradeoff)**.** *Let $f^* \in \mathcal{H}^s_\mu(R)$ and suppose we observe $y_i = f^*(x_i) + \varepsilon_i$, $i = 1, \dots, n$, with $x_i \sim \mu$ i.i.d. and ε_i i.i.d. sub-Gaussian with parameter σ^2. Consider the truncated spectral estimator*

$$\widehat{f}_k(x) = \sum_{j=1}^{k} \widehat{f}_j^{(n)} \, \phi_j(x), \qquad \widehat{f}_j^{(n)} = \frac{1}{n} \sum_{i=1}^{n} y_i \, \phi_j(x_i).$$

Then the expected risk decomposes as

$$\mathbb{E}\big[\|\widehat{f}_k - f^*\|^2_{L^2(\mu)}\big] = \underbrace{\sum_{j>k} (\widehat{f}^*_j)^2}_{\mathrm{bias}^2(k)} + \underbrace{\frac{\sigma^2 \, k}{n}}_{\mathrm{variance}(k)} .$$

Under Effective Dimension Hypothesis (Assumption 4.1) ($\lambda_j \asymp j^{2/d_{\mathrm{eff}}}$) and the Sobolev constraint $\sum_j \lambda_j^s (\widehat{f}^*_j)^2 \le R^2$:

Bias bound. *The tail energy is controlled by the smoothness and dimension:*

$$\mathrm{bias}^2(k) = \sum_{j>k} (\widehat{f}^*_j)^2 \le R^2 \, \lambda_{k+1}^{-s} \asymp R^2 \, k^{-2s/d_{\mathrm{eff}}}.$$

Variance bound.

$$\mathrm{variance}(k) = \frac{\sigma^2 \, k}{n}.$$

Optimal truncation. *Balancing bias and variance:*

$$k^{-2s/d_{\mathrm{eff}}} \asymp \frac{k}{n} \qquad \Longrightarrow \qquad k^* \asymp n^{d_{\mathrm{eff}}/(2s+d_{\mathrm{eff}})}.$$

Minimax rate. *Substituting k^* into either term:*

$$\boxed{\mathbb{E}\big[\|\widehat{f}_{k^*} - f^*\|^2_{L^2(\mu)}\big] \;\lesssim\; n^{-2s/(2s+d_{\mathrm{eff}})}.}$$

This rate is minimax optimal over $\mathcal{H}^s_\mu(R)$.

Remark 4.3 (Interpretation)**.** The three quantities s, d_{eff}, and n interact as follows:

- **Smoothness** s: higher s means faster eigenvalue decay of the coefficients $\widehat{f}^*_j$, so the bias decreases faster with k and fewer modes suffice.
- **Effective dimension** d_{eff}: larger d_{eff} means eigenvalues $\lambda_j \asymp j^{2/d_{\mathrm{eff}}}$ grow more slowly, so more modes carry significant energy and the variance cost of including them is higher. This is the curse of dimensionality in d_{eff}, not in the ambient dimension.
- **Optimal number of modes** k^*: this is the resolution at which you can reliably estimate the function given n samples. Modes $j \le k^*$ are estimated; modes $j > k^*$ are discarded as noise.

Theorem 4.4 (Minimax rate over Besov balls **?**)**.** *Let $f \in \mathcal{B}^s_{p,q}(R) \subset L^r(\mathbb{R}^d)$ with $s > d(1/p - 1/r)_+$. Then the minimax L^r estimation rate from n noisy samples is*

$$\inf_{\widehat{f}_n} \sup_{f \in \mathcal{B}^s_{p,q}(R)} \mathbb{E}\big[\|\widehat{f}_n - f\|_{L^r}\big] \asymp \begin{cases} n^{-\frac{s}{2s+d}}, & \text{if } p \ge r \quad \text{(dense zone)}, \\ \left(\dfrac{\log n}{n}\right)^{\frac{s-d(1/p-1/r)}{2(s-d(1/p-1/r))+d}}, & \text{if } p < r \quad \text{(sparse zone)}. \end{cases}$$

The dense rate is achieved by linear methods (projection, kernel smoothing). The sparse rate requires nonlinear wavelet thresholding; linear estimators are suboptimal by a polynomial factor in $\log n$.

4.2 Compression

4.3 Flow Matching

4.4 Transformers

4.5 ASTs

Definition 4.5 (AND–OR Abstract Syntax Tree)**.** Let $\mathcal{X}$ be an input space and $\phi : \mathcal{X} \to \mathbb{R}^d$ an embedding. An *AND–OR abstract syntax tree (AST)* is defined recursively as:

- **Atom (Leaf):**
$$T(x) = g(x), \qquad g \in \mathcal{G},$$
where $\mathcal{G}$ is a class of simple local models (e.g. linear or affine functions).
- **AND node:**
$$T(x) = \mathbb{I}\big[C_1(x) \wedge \cdots \wedge C_k(x)\big] \cdot T'(x),$$
where each $C_i(x)$ is a constraint and T' is a subtree.
- **OR node:**
$$T(x) = \sum_{j=1}^{m} \mathbb{I}\big[C_j(x)\big] \cdot T_j(x),$$
where $\{C_j\}_{j=1}^m$ form a partition of $\mathcal{X}$ and T_j are subtrees.

The constraints $C(x)$ are typically hyperplane tests of the form
$$C(x) := \langle w, \phi(x) \rangle \geq 0.$$

Semantics.

- AND nodes represent conjunctions of constraints and correspond to intersections of half-spaces or cross-products of independent factors.
- OR nodes represent partitions of the input space into disjoint regions.
- Each root-to-leaf path defines a region
$$\mathcal{X}_\ell = \bigcap_{C \in \text{path}(\ell)} C,$$
on which the function reduces to a simple local model.

Example: 1D Piecewise Linear Function Let
$$f(x) = \begin{cases} a_1 x + b_1, & x \leq t, \\ a_2 x + b_2, & x > t. \end{cases}$$

This function admits the AND–OR representation
$$f(x) = \mathbb{I}[x \leq t]\,(a_1 x + b_1) + \mathbb{I}[x > t]\,(a_2 x + b_2).$$

Equivalently, as an AST:
$$\texttt{OR}\Big(\texttt{AND}(x \leq t,\ \texttt{LINEAR}(a_1, b_1)), \texttt{AND}(x > t,\ \texttt{LINEAR}(a_2, b_2))\Big).$$

Example: Hierarchical Partition Let $t_1 < t_2$ and
$$f(x) = \begin{cases} a_1 x + b_1, & x \leq t_1, \\ a_2 x + b_2, & t_1 < x \leq t_2, \\ a_3 x + b_3, & x > t_2. \end{cases}$$

Then f can be written as
$$f(x) = \mathbb{I}[x \leq t_2]\Big(\mathbb{I}[x \leq t_1](a_1 x + b_1) + \mathbb{I}[t_1 < x \leq t_2](a_2 x + b_2)\Big) + \mathbb{I}[x > t_2](a_3 x + b_3).$$

This corresponds to a depth–2 AND–OR tree.

Example: ReLU as AND–OR For $w \in \mathbb{R}^d$,

$$\mathrm{ReLU}(\langle w, x \rangle) = \mathbb{I}[\langle w, x \rangle \geq 0]\langle w, x \rangle + \mathbb{I}[\langle w, x \rangle < 0] \cdot 0.$$

Thus a ReLU unit is an OR node with two AND branches. A finite ReLU network corresponds to a (possibly large) AND–OR tree, whose leaves are linear functions and whose OR structure encodes the activation pattern partition.

[Emile says: What kind of non-trivial theoretical results to validate this construction would be the most insightful? One thing is that for coarse/non-granular functions, it should sort of recover the original manifold hypothesis? What are the theoretical results pertaining to manifold hypothesis in the other papers we've been reading? Can we establish any analogs?]

[Emile says: What is the simplest possible way to empirically validate anything that we've constructed (after constructing it)?]

A Learning the regularizer

Assume, the regularizer depends on some parameter θ. Then learning it is equivalent to minimizing the negative log-likelihood:

$$\min_\theta \mathcal{L}(\theta) = \lambda \int_f R_\theta(f) d\pi(f) + \log Z(\theta)$$

where $Z(\theta) = \int_{f\in X} \exp(-R_\theta(f))\nu(df)$ is a normalizing constant, X is function space, ν base reference measure on X.

Or, if we are given the empirical observations of functions $\{f_i\}_{i=1}^t$ then

$$\min_\theta \mathcal{L}(\theta) = \frac{\lambda}{t}\sum_i R_\theta(f_i) + \log Z(\theta)$$

Assuming $R_\theta(f) = \langle \theta, \Phi(f)\rangle$ (linear in θ) out problem is convex, yaaaaay! Parameter θ can be here an infinite-dimensional measure or a vector, hence, the minimization problem can be solved by gradient descent on θ.

The gradient of the data term is

$$\nabla_\theta \left(\frac{1}{T}\sum_{t=1}^T \langle \theta, \Phi(f^{(t)})\rangle\right) = \frac{1}{T}\sum_{t=1}^T \Phi(f^{(t)}) =: \mathbb{E}_{\text{data}}[\Phi(f)]\,.$$

For the partition function term, differentiation under the integral yields

$$\begin{aligned}\nabla_\theta \log Z(\theta) &= \frac{1}{Z(\theta)}\nabla_\theta Z(\theta)\\ &= \frac{1}{Z(\theta)}\int (-\Phi(f))\exp\big(-\langle\theta,\Phi(f)\rangle\big)\,\nu(df)\\ &= -\mathbb{E}_{f\sim p_\theta}[\Phi(f)]\,.\end{aligned}$$

Combining the two terms, the gradient of the objective is

$$\boxed{\nabla_\theta\mathcal{L}(\theta) = \mathbb{E}_{\text{data}}[\Phi(f)] - \mathbb{E}_{f\sim p_\theta}[\Phi(f)]\,,}.$$

A gradient descent step with step size $\eta > 0$ is therefore given by

$$\boxed{\theta_{k+1} = \theta_k - \eta\left(\mathbb{E}_{\text{data}}[\Phi(f)] - \mathbb{E}_{f\sim p_{\theta_k}}[\Phi(f)]\right).}$$

If θ is constrained (e.g., $\theta \geq 0$ or θ lies in a simplex), the update is followed by projection onto the feasible set Θ:

$$\theta_{k+1} = \Pi_\Theta\left[\theta_k - \eta\left(\mathbb{E}_{\text{data}}[\Phi(f)] - \mathbb{E}_{f\sim p_{\theta_k}}[\Phi(f)]\right)\right].$$

At a stationary point $\theta^\star$, the optimality condition is the moment-matching identity

$$\mathbb{E}_{\text{data}}[\Phi(f)] = \mathbb{E}_{f\sim p_{\theta^\star}}[\Phi(f)]\,.$$

$\Phi(f)$ here a statistic extractor on function space.

B Examples

Note, in all the examples that follow $\Phi(f)$ is convex in f.

Example 1 (Energy regularization).

$$\Phi : L^2(M) \to \mathbb{R}, \qquad \Phi(f) := \|f\|^2_{L^2(M)}.$$

The induced regularizer $R(f) = \|f\|^2_{L^2}$ corresponds to ridge/Tikhonov regularization. This choice measures only the magnitude of f and ignores geometric structure.

B.1 First-Order Smoothness

Example 2 (Sobolev H^1 regularization).

$$\Phi : H^1(M) \to \mathbb{R}, \qquad \Phi(f) := \int_M \|\nabla f(x)\|^2 \, dx.$$

This feature measures global smoothness. The null space consists of constant functions.

Example 3 (Anisotropic Sobolev regularization). Let $\{Q_j\}_{j=1}^k$ be fixed symmetric positive semidefinite matrices. Define

$$\Phi : H^1(M) \to \mathbb{R}^k, \qquad \Phi_j(f) := \int_M \langle \nabla f(x), Q_j \nabla f(x) \rangle \, dx.$$

Then

$$R_\theta(f) = \sum_{j=1}^k \theta_j \Phi_j(f)$$

corresponds to learning a Riemannian metric.

B.2 Total Variation and Sparsity

Example 4 (First-order total variation).

$$\Phi : BV(M) \to \mathbb{R}_+, \qquad \Phi(f) := \|\nabla f\|_{\mathcal{M}}.$$

This feature measures the total mass of the gradient as a Radon measure and promotes piecewise-constant functions.

B.3 Second-Order and Curvature-Based Statistics

Example 5 (Second-order total variation / splines).

$$\Phi : X \to \mathbb{R}_+, \qquad \Phi(f) := \|\Delta f\|_{\mathcal{M}},$$

where X is the space of functions with measure-valued Laplacian.

The null space consists of affine functions, and minimizers are adaptive splines.

Example 6 (Anisotropic second-order TV). Let $\mathcal{S}_+$ denote the cone of symmetric positive semidefinite matrices. Define

$$\Phi : X \to \mathcal{C}(\mathcal{S}_+, \mathbb{R}_+), \qquad \Phi(f)(Q) := \|\Delta_Q f\|_{\mathcal{M}},$$

where $\Delta_Q f := \nabla \cdot (Q \nabla f)$.

Here:

- domain: functions with measure-valued anisotropic Laplacian,
- codomain: nonnegative functions on $\mathcal{S}_+$,
- regularizer:

$$R_\theta(f) = \int_{\mathcal{S}_+} \|\Delta_Q f\|_{\mathcal{M}} \, d\theta(Q).$$

This promotes sparse, geometry-aligned curvature and induces a Finsler total variation.

B.4 Spectral Statistics

Example 7 (Spectral energy features). Let $\{\psi_k\}$ be eigenfunctions of a Laplacian on M. Define

$$\Phi : L^2(M) \to \ell^1, \qquad \Phi_k(f) := |\langle f, \psi_k \rangle|^2.$$

The corresponding regularizer weights energy by frequency and induces a spectral bias.

B.5 Measure-Based and Neural Representations

Example 8 (Barron / ReLU representation). Let

$$f_\mu(x) = \int_{\mathbb{S}^{d-1}} \sigma(\langle w, x\rangle)\, d\mu(w),$$

with μ a finite signed Radon measure. Define

$$\Phi : \mathcal{B} \to \mathcal{M}_+(\mathbb{S}^{d-1}), \qquad \Phi(f_\mu) := |\mu|.$$

Here:

- domain: Barron space $\mathcal{B}$,
- codomain: finite nonnegative measures,
- regularizer: $R(f_\mu) = \|\mu\|_{\mathrm{TV}}$.

This promotes sparse ridge representations.

Example 9 (Finsler–TV / Radon–Laplacian). Combining the previous examples,

$$\Phi : \mathcal{B} \to \mathcal{C}(\mathcal{S}_+, \mathbb{R}_+), \qquad \Phi(f_\mu)(Q) := \int \sqrt{w^\top Q w}\, |\mu|(dw).$$

Equivalently,

$$R(f_\mu) = \int_{\mathcal{S}_+} \Phi(f_\mu)(Q)\, d\theta(Q) = \int F(w)\, |\mu|(dw), \quad F(w) := \int \sqrt{w^\top Q w}\, d\theta(Q).$$

This is a Finsler total variation norm on kink directions.

C Back to the problem

Given all distribution over functions minimizing the negative log-likelihood is

$$\min_\theta \mathcal{L}(\theta) = \lambda \int_f R_\theta(f) d\pi(f) + \log Z(\theta)$$

where $Z(\theta) = \int_{f\in X} \exp(-R_\theta(f))\nu(df)$ is a normalizing constant, X is function space, ν base reference measure on X.

However, everything we observe is just a discrete sample. Hence, given the empirical observations of functions $\{f_i\}_{i=1}^t$ our minimization looks like

$$\min_\theta \mathcal{L}(\theta) = \frac{\lambda}{t} \sum_i R_\theta(f_i) + \log Z(\theta)$$

Again, **everything** we observe is a discrete sample. Usually, we have access to each function f via its values: $D = \{(x_i, y_i)\}$ where $y_i = f(x_i) + \mathcal{N}(0, \sigma^2)$. Note, that

$$\log p_\theta(D_Y | f, D_X) \propto -\frac{1}{\sigma^2} \sum_{(x,y)\in D} \|y - f(x)\|_2^2 = \ell(D, f)$$

Hence,

$$-\log p_\theta\left(D_Y \mid D_X\right) = -\log \int \exp\left(-\ell(D, f) - \lambda R_\theta(f)\right) \nu(df) + \log Z(\theta)$$

So given datasets $\{D^{(i)}\}_{i=1}^t$ our optimization problem is

$$\min_\theta \mathcal{L}(\theta) = -\frac{1}{t} \sum_{i=1}^t \log \int \exp\left(-\ell\left(D^{(i)}, f\right) - \lambda R_\theta(f)\right) \nu(df) + \log Z(\theta) \tag{1}$$

However, the term $\log \int \exp \ldots$ is intractable :(This is why the masking comes into play in practice.

D Masking

D.1 Posterior concentration around the MAP estimator

Fix a parameter θ and a dataset $D = \{(x_j, y_j)\}_{j=1}^{n}$. Define the energy functional

$$E_\theta(f) := \ell(D, f) + \lambda R_\theta(f),$$

where $\ell(D, f)$ denotes the empirical negative log-likelihood and R_θ is a regularizer.

Assume the following:

1. The dataset D is large (large n) and noise variance σ^2 is small
2. (*Coercivity*) The functional R_θ is coercive on X modulo a finite-dimensional nullspace $\mathcal{N}$, i.e., $\|f\|_X \to \infty$ with $f \perp \mathcal{N}$ implies $R_\theta(f) \to \infty$.
3. (*Strict convexity*) The functional $\ell(D, \cdot)$ is strictly convex on $X/\mathcal{N}$.
4. (*Regularity*) The map $f \mapsto E_\theta(f)$ is twice differentiable in a neighborhood of its minimizer.

Under these assumptions, the minimization problem

$$\widehat{f}_\theta \in \arg\min_{f \in X} E_\theta(f)$$

admits a unique solution modulo $\mathcal{N}$. Moreover, the posterior distribution

$$p_\theta(f \mid D) \propto \exp\big(-E_\theta(f)\big)$$

concentrates in probability around $\widehat{f}_\theta$ as either $n \to \infty$ or $\sigma^2 \to 0$. In particular, for any neighborhood U of $\widehat{f}_\theta$ in X,

$$p_\theta(U \mid D) \;\longrightarrow\; 1.$$

D.2 Laplace approximation of the marginal likelihood

Let $E_\theta(f) = \ell(D, f) + \lambda R_\theta(f)$ and let $\widehat{f}_\theta$ denote its unique minimizer. Define the Hessian operator

$$H_\theta := \nabla^2 E_\theta(\widehat{f}_\theta),$$

acting on the tangent space of $X/\mathcal{N}$.

By a second-order Taylor expansion,

$$E_\theta(f) = E_\theta(\widehat{f}_\theta) + \frac{1}{2}\langle f - \widehat{f}_\theta,\, H_\theta(f - \widehat{f}_\theta)\rangle + o(\|f - \widehat{f}_\theta\|_X^2).$$

Substituting this expansion into the marginal likelihood integral yields the Laplace approximation

$$\int_X \exp\big(-E_\theta(f)\big)\,\nu(df) \;\approx\; \exp\big(-E_\theta(\widehat{f}_\theta)\big) \int \exp\big(-\tfrac{1}{2}\langle h, H_\theta h\rangle\big)\; dh. \tag{2}$$

The remaining integral is Gaussian and evaluates (up to a multiplicative constant) to $(\det H_\theta)^{-1/2}$, where $\det H_\theta$ denotes a suitable regularized determinant. Consequently,

$$\int_X \exp\big(-\ell(D, f) - \lambda R_\theta(f)\big)\,\nu(df) \;\approx\; \exp\big(-\ell(D, \widehat{f}_\theta) - \lambda R_\theta(\widehat{f}_\theta)\big)(\det H_\theta)^{-1/2}. \tag{3}$$

D.3 Decomposition of the marginal likelihood objective

Substituting the Laplace approximation (3) into the negative log marginal likelihood yields

$$-\log p_\theta(D_Y \mid D_X) \;\approx\; \ell(D, \widehat{f}_\theta) + \lambda R_\theta(\widehat{f}_\theta) + \frac{1}{2}\log\det H_\theta - \log Z(\theta), \tag{4}$$

up to additive constants independent of θ.

This expression decomposes the marginal likelihood into four interpretable components:

1. *Data-fit term* $\ell(D, \widehat{f}_\theta)$, measuring how well the MAP estimator explains the observations;
2. *Regularization cost* $\lambda R_\theta(\widehat{f}_\theta)$, encoding alignment between the learned function and the inductive bias imposed by θ;
3. *Local complexity (Occam) penalty* $\frac{1}{2}\log\det H_\theta$, which penalizes flat or unstable directions around the MAP solution;
4. *Global complexity penalty* $-\log Z(\theta)$, measuring the overall volume of functions favored by the prior.

Together, these terms characterize the tradeoff between data fit, regularization, and model complexity in the marginal likelihood objective.

E Back to the optimization

Given the collection of datasets $\mathcal{D} = \left\{\left(D^{(i)}_{\text{train}}, D^{(i)}_{\text{test}}\right)\right\}_{i=1}^{t}$ Substituting this into Equation (1) we have

$$\min_\theta \mathcal{L}(\theta) = \frac{1}{t}\sum_{i=1}^{t} \ell\left(D^{(i)}_{\text{test}}, \widehat{f}^{(i)}_\theta\right) + \lambda R_\theta\left(\widehat{f}^{(i)}_\theta\right) + \frac{1}{2}\log\det H_\theta - \log Z(\theta)$$
$$\text{where } \widehat{f}^{(i)}_\theta = \operatorname*{argmin}_f \ell\left(D^{(i)}_{\text{train}}, f\right) + \lambda R_\theta\left(f\right)$$

This is bilever optimization. Even though it lost the convexity, it is tractable and can be solved in practice by alternating gradient descent. I claim that gradient descent of the outer problem = training a transformer, gradient descent of the inner problem is a transformer forward pass.

F Inner problem

For a dataset $D = \{(x_i, y_i)\}$ we have

$$\widehat{f}_\theta = \operatorname*{argmin}_f \ell\left(D, f\right) + \lambda R_\theta\left(f\right) = \operatorname*{argmin}_f \|f(x_i) - y_i\|_2^2 + \lambda\langle\theta, \Phi(f)\rangle$$

Note, that if Φ is convex in f then this can be solved by subgradient descent on f:

$$f^{(t+1)} = f^{(t)} - \eta\left[2\sum_i \big(f^{(t)}(x_i) - y_i\big)\delta_{x_i} + \lambda\, g^{(t)}\right]$$

where $g^{(t)} \in \partial_f R_\theta(f^{(t)})$ is a subgradient of the regularizer.

F.1 How transformer forward pass can implement a (sub)gradient descent

Now lets go through some of the examples in Appendix B.3 to see how transformer-like networks can implement a (sub)gradient g descent for the inner problem.

Energy regularization

$$\Phi : L^2(M) \to \mathbb{R}, \qquad \Phi(f) := \|f\|^2_{L^2(M)} \qquad R_\theta(f) = \theta\|f\|^2_{L^2(M)} \qquad g = 2f$$

Anisotropic Sobolev Regularization Parameter θ is a measure over PSD matrices.

$$\Phi(f)(Q) := \|\nabla f(x)\|^2_Q \qquad R_\theta(f) = \int_{Q\in\mathbf{S}_+} \|\nabla f(x)\|^2_Q d\theta(Q) \qquad g = -\int_{Q\in\mathbf{S}_+} \Delta_Q f d\theta(Q)$$

Anisotropic second-order TV

$$\Phi : X \to \mathcal{C}(\mathcal{S}_+, \mathbb{R}_+), \qquad \Phi(f)(Q) := \|\Delta_Q f\|_{\mathcal{M}}, \qquad g = \int_{\mathcal{S}_+} \Delta_Q \operatorname{sign}(\Delta_Q f) d\theta(Q)$$

where $\Delta_Q f := \nabla \cdot (Q \nabla f)$.

As the functions are defined on a "flat" domain $\mathcal{M} \subset R^d$, the discrete version of Δ_Q looks like:

$$L \in R^{n \times n} : L_{ij} = \exp\left(\frac{-\|x_i - x_j\|^2_{Q^{-1}}}{\epsilon}\right)$$

If our domain is curved manifold $\mathcal{M}$ we can W.L.O.G. (by Takahashi's thm hehe) assume that it's $\mathcal{M} \in S^{d-1}$, hence, the discrete version of Δ_Q will require projection $P_i = I - x_i x_i^\top$:

$$\begin{aligned} L \in R^{n \times n} : L_{ij} &= \exp\left(\frac{-(x_i - x_j)^\top P_i Q^{-1} P_i (x_i - x_j)^\top}{\epsilon}\right) \\ &= \exp\left(-\frac{\left(x_j - \left(x_i^\top x_j\right) x_i\right)^\top Q^{-1} \left(x_j - \left(x_i^\top x_j\right) x_i\right)}{\epsilon}\right) \end{aligned}$$

Note, that all the exp terms are kind of similar to the original transformer softmax terms. I think, softmax should be replaced by those. The final idea is that a transformish NN with infinite amount of heads approximates the distribution over Q.

TODO: move this

Theorem F.1 (Stability from coercive Φ-regularization)**.** *Let $(X, \|\cdot\|_X)$ be a reflexive Banach space of functions on a domain M, and let*

$$R_\theta(f) := \langle \theta, \Phi(f) \rangle \in [0, \infty]$$

where $\Phi : X \to V$ is a measurable map into a real locally convex space V and $\theta \in V^*$ is such that R_θ is a proper, convex, lower semicontinuous functional on X. Assume the following:

1. ***Coercivity (modulo a nullspace).*** *There exists a finite-dimensional subspace $\mathcal{N} \subset X$ and a constant $c > 0$ such that for all $f \in X$,*

$$R_\theta(f) \;\geq\; c\,\|f - \Pi_{\mathcal{N}} f\|_X - c,$$

where $\Pi_{\mathcal{N}}$ denotes a fixed linear projection onto $\mathcal{N}$. In particular, the sublevel sets $\{f : R_\theta(f) \leq C\}$ are relatively compact in $X/\mathcal{N}$.

2. ***Strong convexity of the empirical loss.*** *For each dataset $D = \{(x_i, y_i)\}_{i=1}^n$, define*

$$\mathcal{L}_D(f) := \frac{1}{n} \sum_{i=1}^{n} \ell\big(f(x_i), y_i\big),$$

and assume $\mathcal{L}_D$ is α-strongly convex on $X/\mathcal{N}$ with respect to $\|\cdot\|_X$, uniformly over all datasets D (for some $\alpha > 0$).

3. ***Lipschitz dependence on function values.*** *Assume $\ell(\cdot, y)$ is L-Lipschitz in its first argument for all y, and that point-evaluation is continuous on X in the sense that there exists $\kappa > 0$ such that for all $x \in M$ and all $f, g \in X$,*

$$|f(x) - g(x)| \leq \kappa \|f - g\|_X.$$

Observed data $y_i = f^*(x_i) + \mathcal{N}(0, \sigma^2)$

$$\widehat{f} = \operatorname*{argmin}_f \frac{1}{n} \sum_{i=1}^{n} \ell(f(x_i), y_i) + \lambda R(f) \Rightarrow \|f^* - \widehat{f}\|_X \leq C(n, \sigma, R, \lambda)$$

Proposition F.2 (Bilevel learning of θ as a MAP surrogate)**.** *Let $R_\theta(f) = \langle \theta, \Phi(f) \rangle$ and define the inner estimator*

$$\widehat{f}_\theta(D) \in \arg\min_{f \in X} \Big\{ \mathcal{L}_D(f) + \lambda R_\theta(f) \Big\}.$$

Consider the bilevel objective based on masking (or train/validation splits),

$$\min_\theta \; \mathbb{E}_{\textit{mask}} \Big[\mathcal{L}_{\mathrm{val}}\big(\widehat{f}_\theta(D_{\mathrm{train}})\big) \Big] + \Omega(\theta).$$

Then bilevel optimization selects θ so as to minimize the predictive risk of the MAP estimator induced by the Gibbs prior

$$p_\theta(df) \propto \exp\big(-\lambda R_\theta(f)\big)\, \nu(df).$$

In contrast, maximum-likelihood learning of θ for the induced Gibbs model includes an additional global complexity term $\log Z(\theta)$, which is generally not captured by the bilevel objective unless explicitly approximated (e.g., via Laplace/Occam corrections).

Proposition F.3 (Bilevel learning of θ as a MAP surrogate)**.** *Let $R_\theta(f) = \langle \theta, \Phi(f) \rangle$ and define the inner estimator*

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